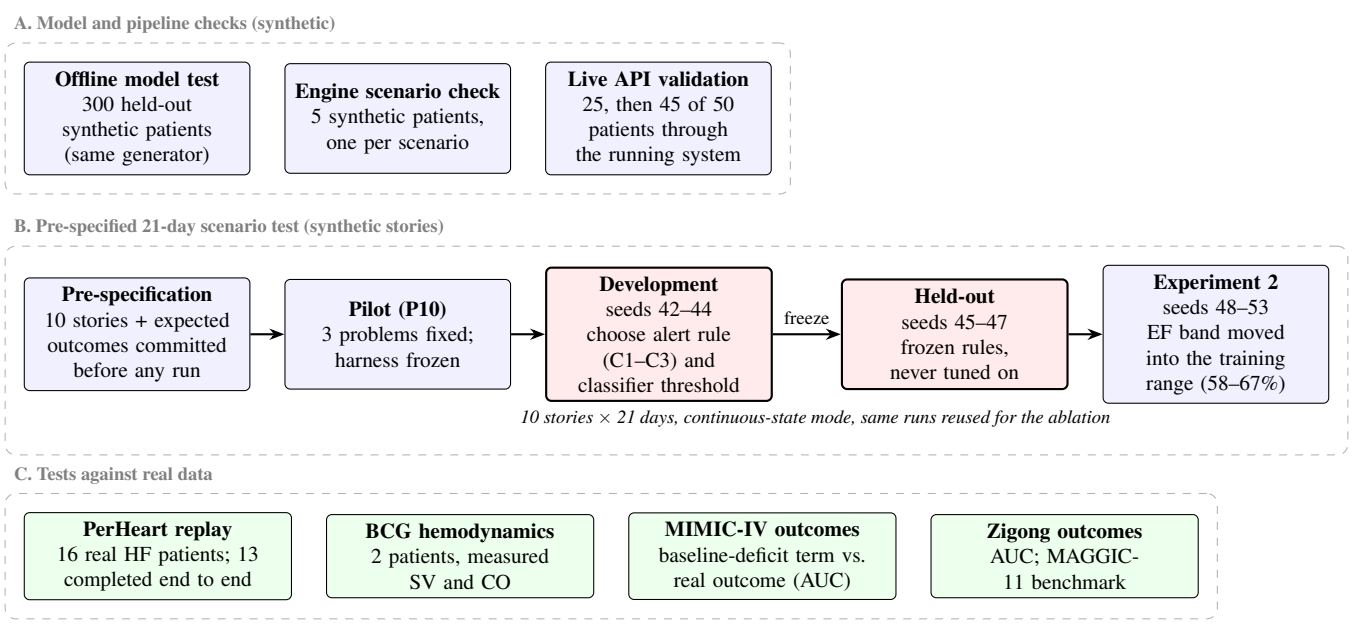
A. Model and pipeline checks (synthetic)
Offline model test
300 held-out synthetic patients (same generator)
Engine scenario check
5 synthetic patients, one per scenario
Live API validation
25, then 45 of 50 patients through the running system
B. Pre-specified 21-day scenario test (synthetic stories)
Pre-specification
10 stories + expected outcomes committed before any run
Pilot (P10)
3 problems fixed; harness frozen
Development
seeds 42–44
choose alert rule (C1–C3) and classifier threshold
freeze
Held-out
seeds 45–47
frozen rules, never tuned on
Experiment 2
seeds 48–53
EF band moved into the training range (58–67%)
10 stories × 21 days, continuous-state mode, same runs reused for the ablation
C. Tests against real data
PerHeart replay
16 real HF patients; 13 completed end to end
BCG hemodynamics
2 patients, measured SV and CO
MIMIC-IV outcomes
baseline-deficit term vs. real outcome (AUC)
Zigong outcomes
AUC; MAGGIC-11 benchmark